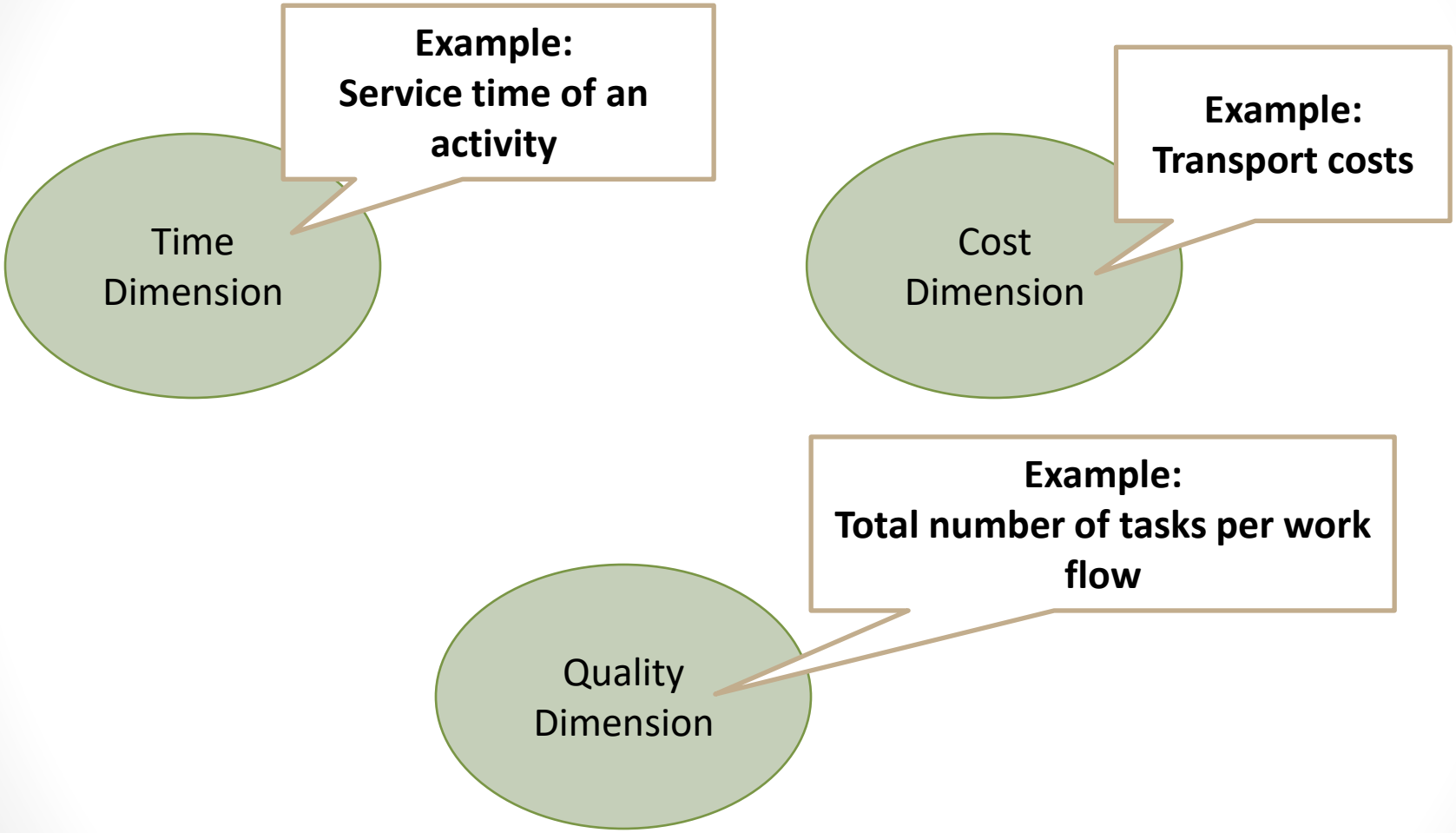
IS341-SIS334

BPM

Extracting Key Performance Indicators from Process Execution Logs

Dimensions of Performance Indicators



Performance Indicators

- **<u>Time Related-Performance Indicators:</u>**
  - Effective Time:
    - Actual time of an object. Object can be activity, human resource.
  - Service Time:
    - Total time of an object.
  - Waiting Time:
    - Idle time of an object.
  - Sojourn time:
    - Time when the activity offered in the system till it terminated.

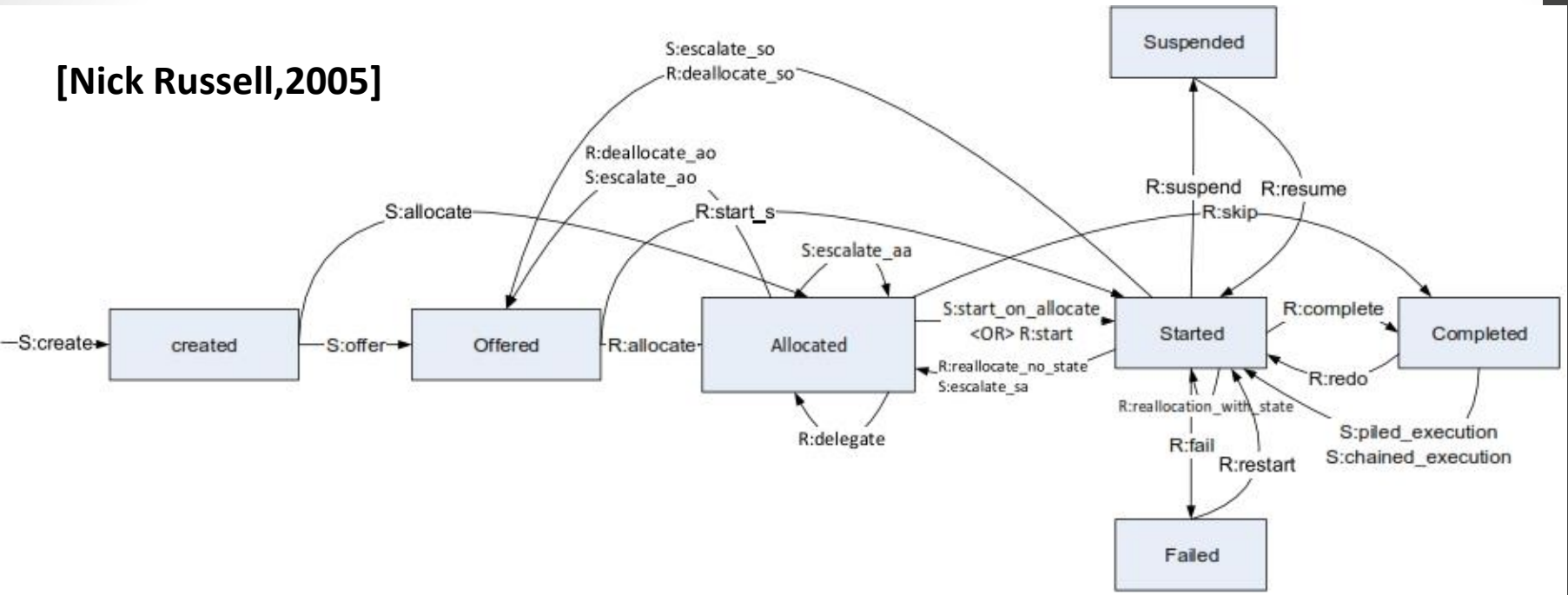
Process Execution logs

- **Process Execution logs** are logs that contain information about the activities, their timestamps and the resources who participated to their execution.

Case	Activity	Resource	Event type	Timestamp
⋮	⋮	⋮	⋮	⋮
1	R.H	Ahmed	Offered	1-6-2021 14:30
1	R.H	Ahmed	Allocated	1-6-2021 15:30
1	R.H	Ahmed	Started	1-6-2021 17:30
1	R.H	Ahmed	Suspended	1-6-2021 23:30
1	R.H	Ahmed	Started	1-6-2021 1:30
1	R.H	Nada	Offered	1-6-2021 2:30
1	R.H	Nada	Allocated	1-6-2021 03:30
1	R.H	Nada	Started	2-6-2021 18:30
1	R.H	Nada	Completed	2-6-2021 23:30
⋮	⋮	⋮	⋮	⋮

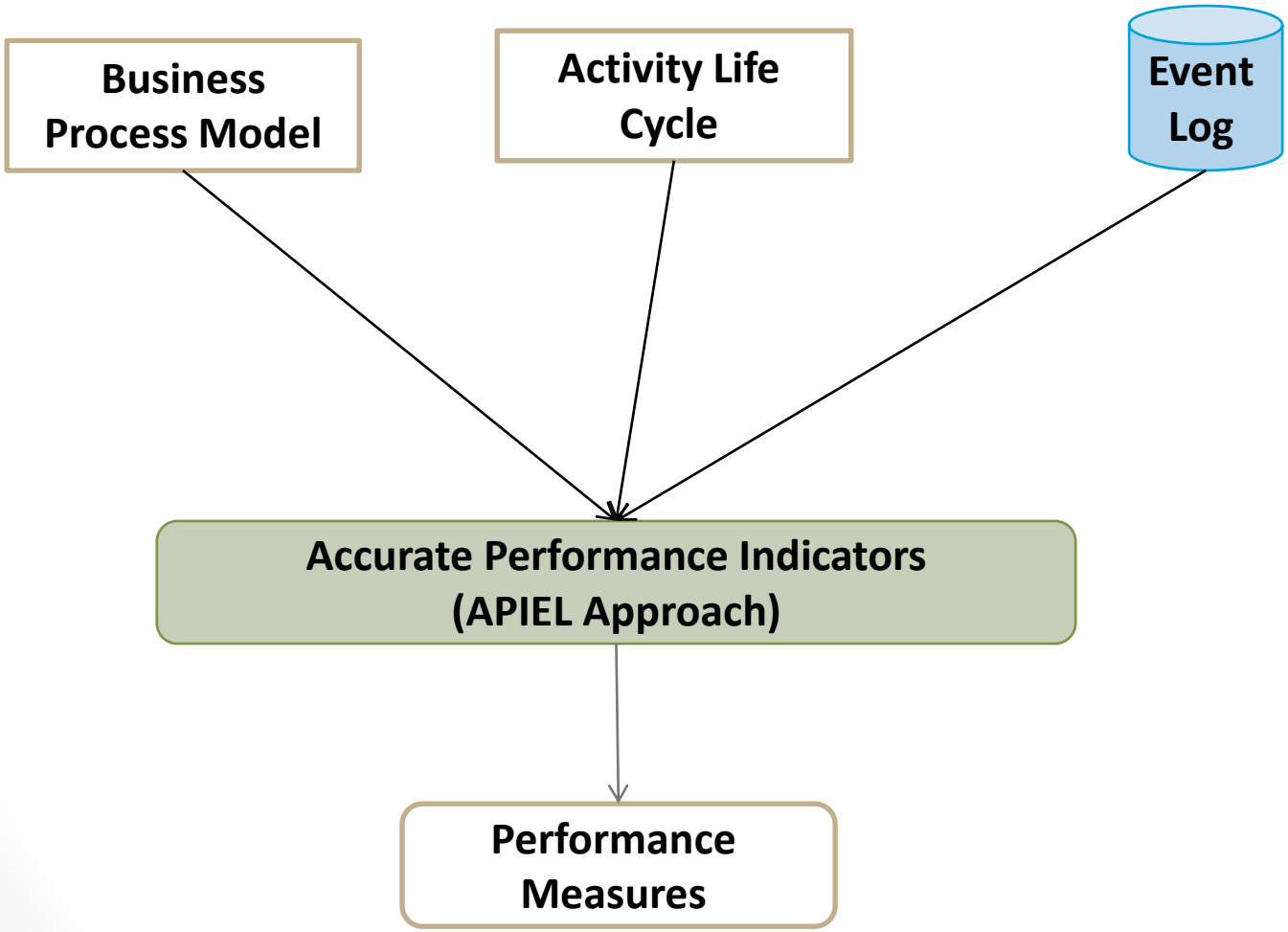
Activity Life Cycle

[Nick Russell,2005]

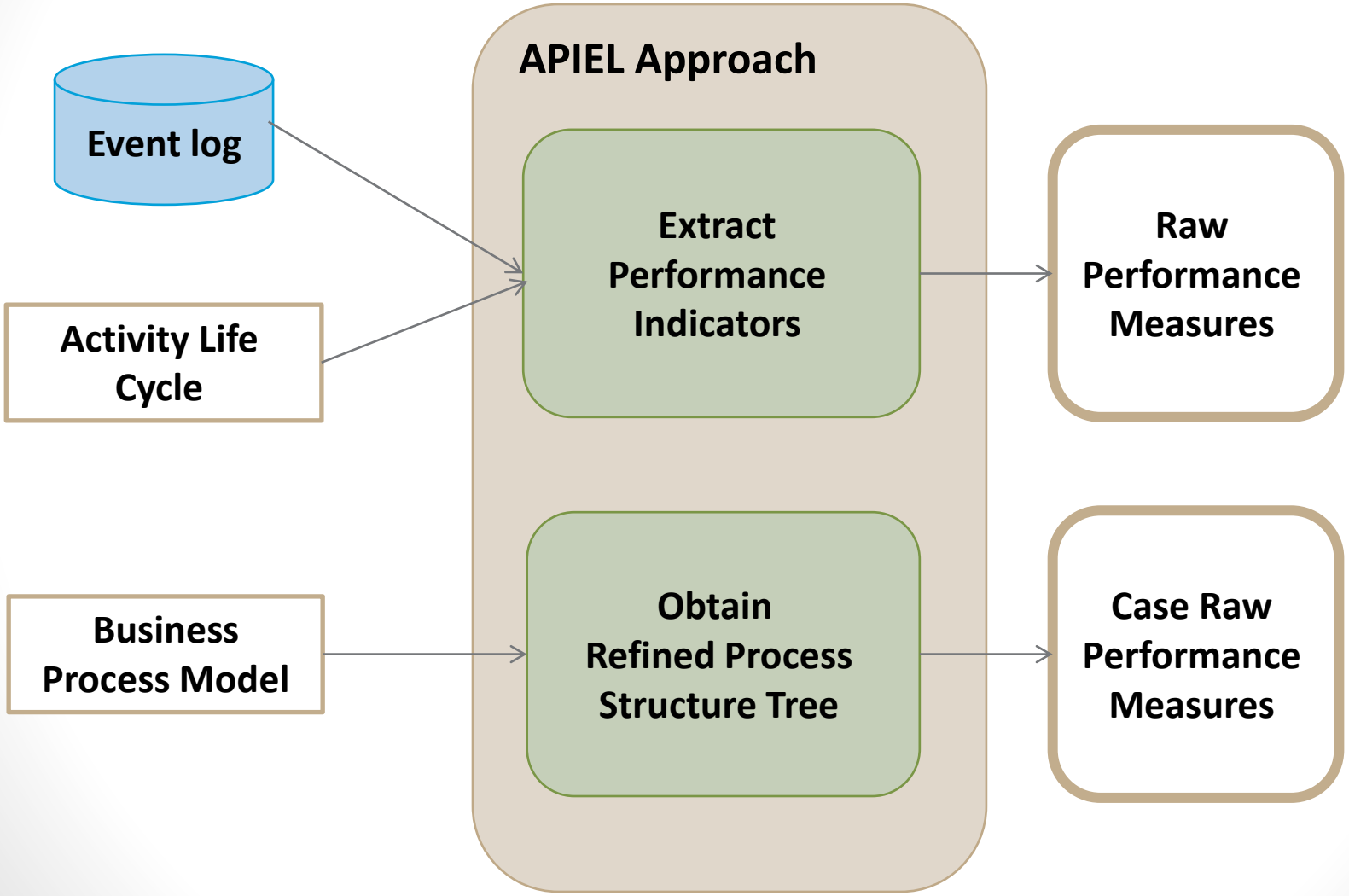


Case ID	Activity	Resource	Event type	TimeStamp
1	A	R1	Offered	1/1/2014 01:00
1	A	R1	Allocated	2/1/2014 02:00
1	A	R1	Started	2/1/2014 04:00
1	A	R1	Completed	4/1/2014 03:00

APIEL approach

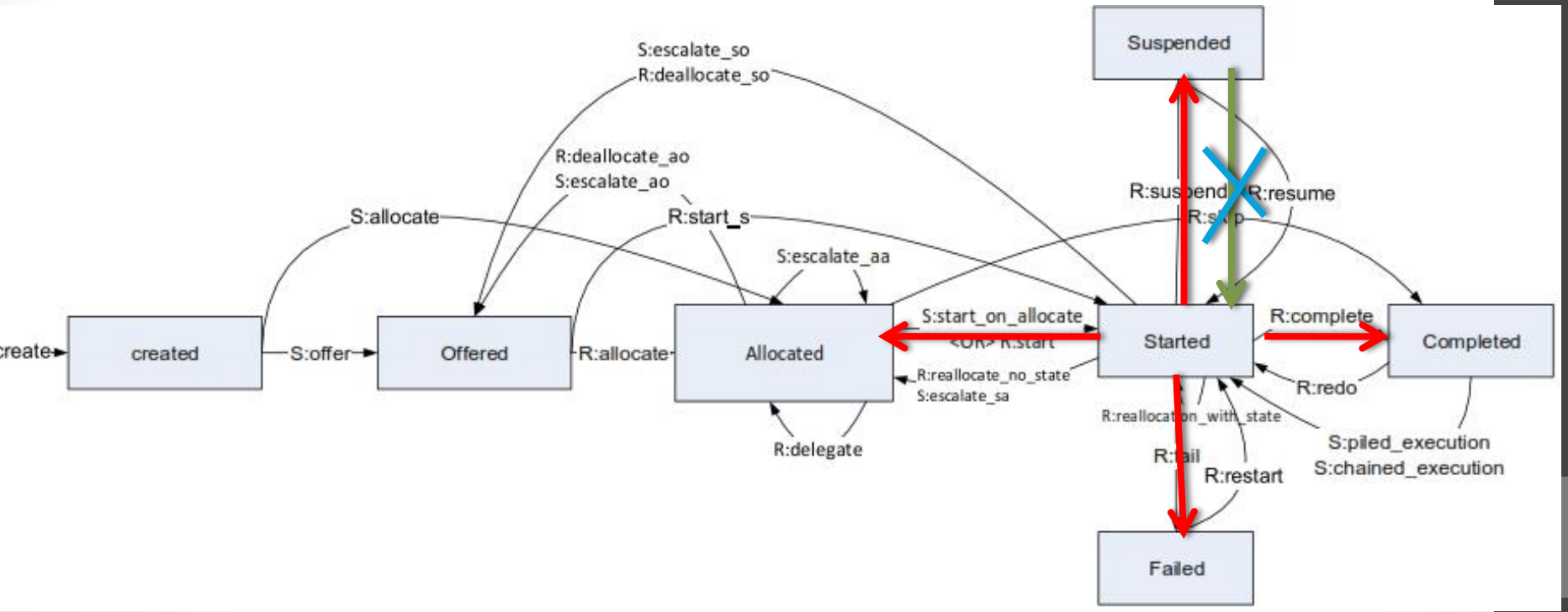


APIEL Approach



Effective Time

- **Effective time** is the actual time that the resource work on the activity without idle time.



Effective Time

- Calculate Effective time of two resources “Ahmed” and “Nada” according to activity Repair Hardware (R.H)within case 1.

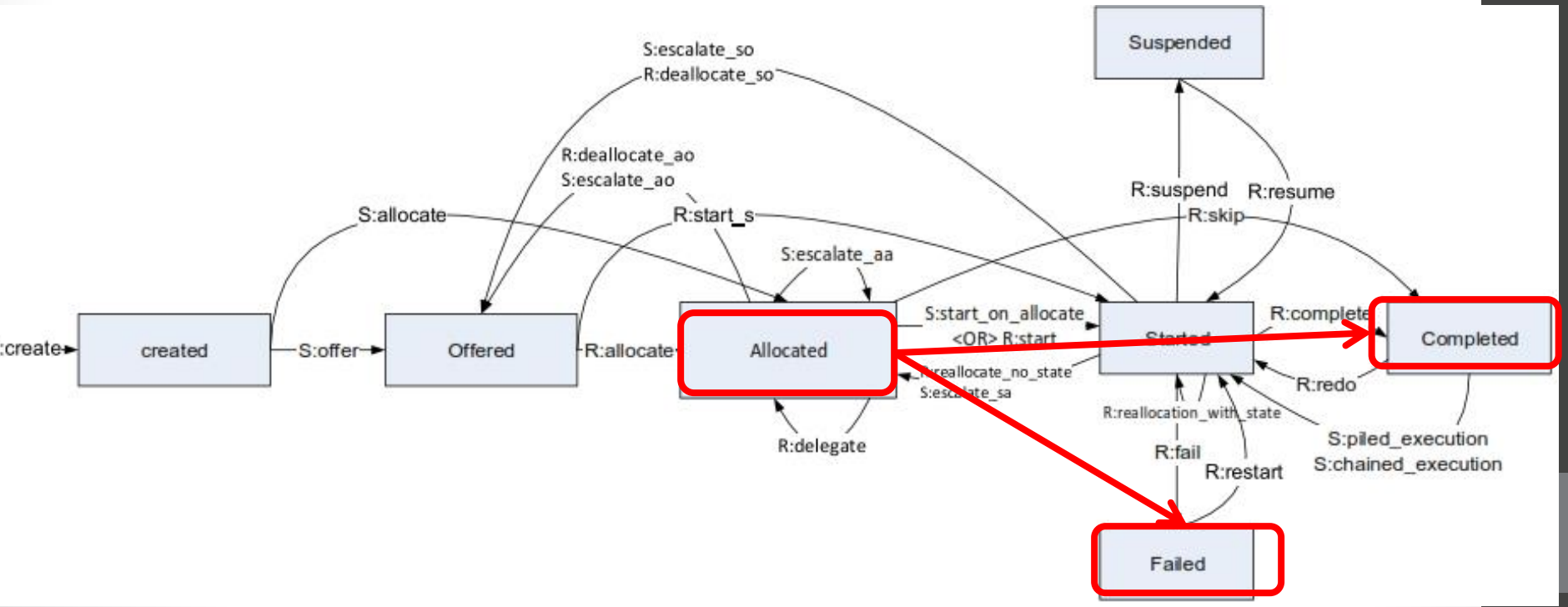
Effective time of Resource “Ahmed” according to Activity “R.H.” within case 1 is **“7 hours”**

Effective time of Resource “Nada” according to Activity “R.H.” within case 1 is **“5 hours”**

Case	Activity	Resource	Event type	Timestamp
⋮	⋮	⋮	⋮	⋮
1	R.H	Ahmed	Offered	1-6-2021 14:30
1	R.H	Ahmed	Allocated	1-6-2021 15:30
1	R.H	Ahmed	Started	1-6-2021 17:30
1	R.H	Ahmed	Suspended	1-6-2021 23:30
1	R.H	Ahmed	Started	1-6-2021 1:30
1	R.H	Nada	Offered	1-6-2021 2:30
1	R.H	Nada	Allocated	1-6-2021 03:30
1	R.H	Nada	Started	2-6-2021 18:30
1	R.H	Nada	Completed	2-6-2021 23:30
⋮	⋮	⋮	⋮	⋮

Service Time

- Is the total time that the resource worked on the activity. The time elapsed between the started time of the activity until it terminated.



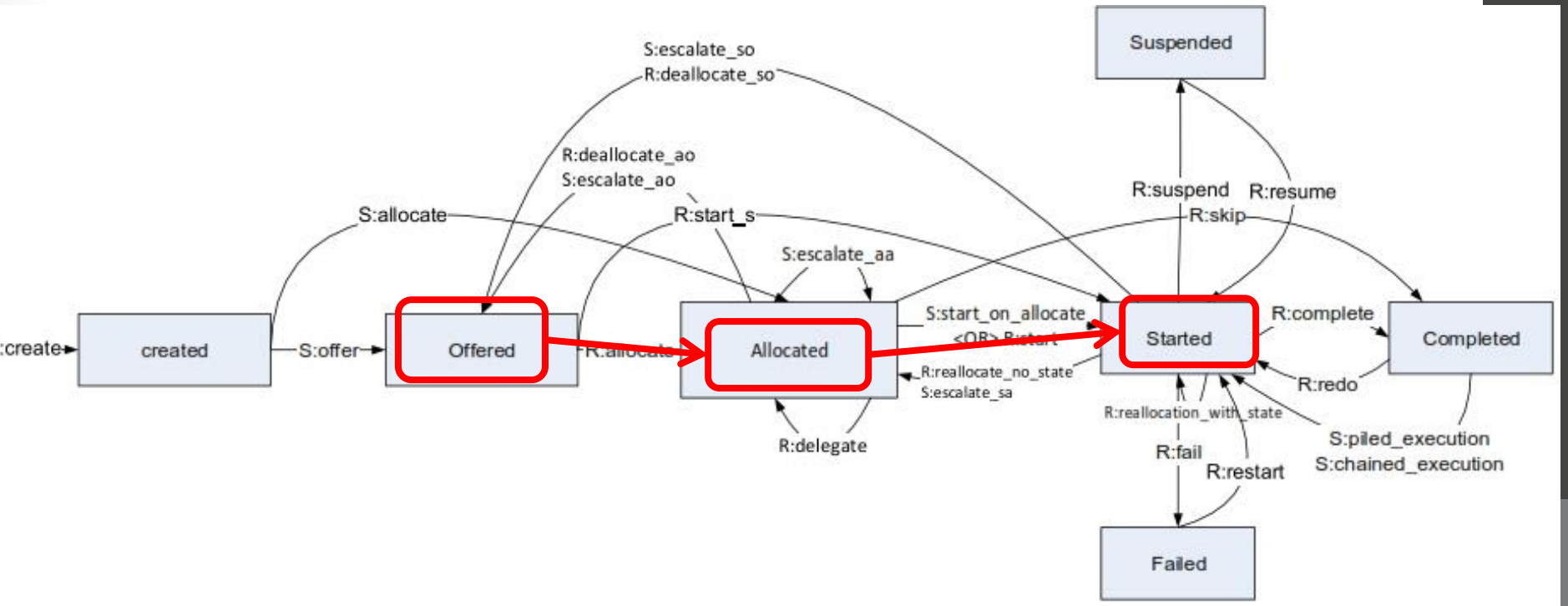
Service Time

- **Example:**
  - Service time for Mohamed in R.S (repair software) in case 2 is 20 hours

Event ID	Case	Activity	Resource	Event type	Timestamp
	⋮	⋮	⋮	⋮	⋮
60	2	R.S	Mohamed	Offered	15-5-2021 13:45
61	2	R.S	Mohamed	Allocated	15-5-2021 14:45
62	2	R.S	Mohamed	Started	15-5-2021 16:45
63	2	R.S	Mohamed	Suspended	15-5-2021 19:45
64	2	R.S	Mohamed	Started	15-5-2021 20:45
65	2	R.S	Mona	Allocated	16-5-2021 10:45
66	2	R.S	Mona	Started	16-5-2021 11:45
67	2	R.S	Mona	Completed	16-5-2021 13:45
	⋮	⋮	⋮	⋮	⋮

Waiting Time

- Time between offered/allocated and started events.



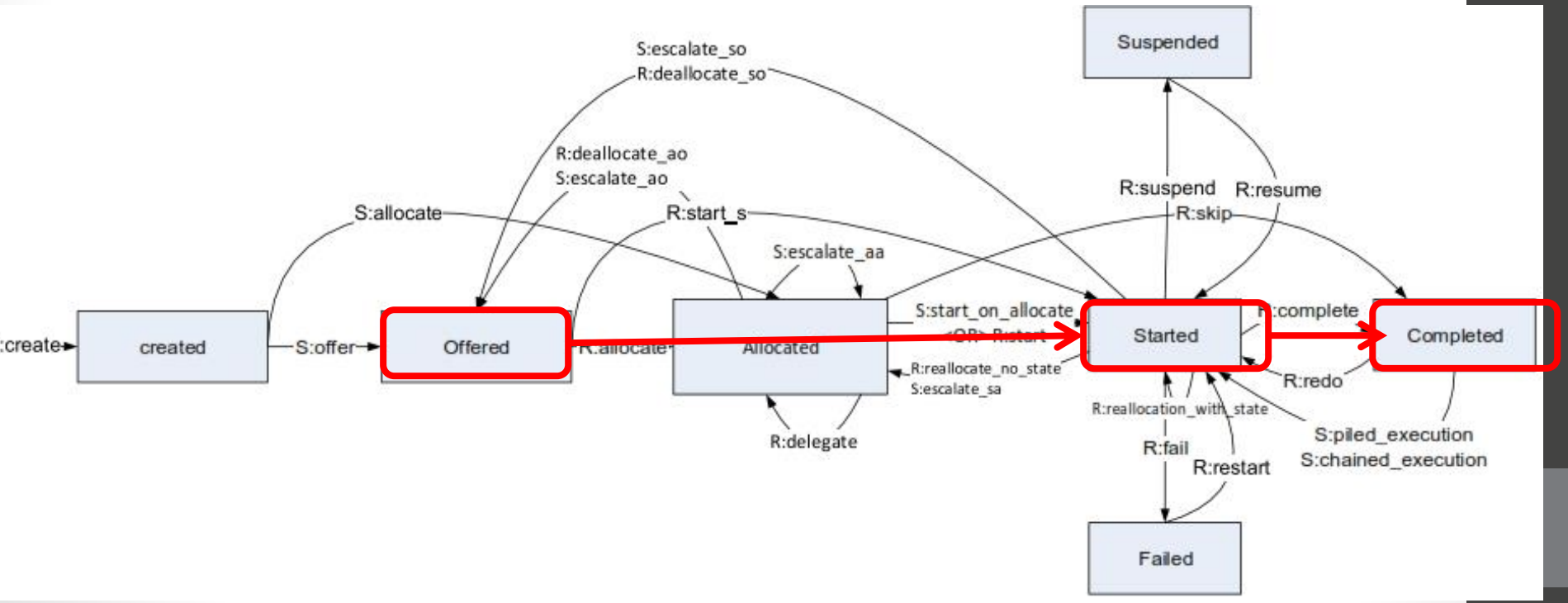
Waiting Time

- **Example:**
  - Waiting time of Ahmed in R.H within case 1 is 2 hours

Case	Activity	Resource	Event type	Timestamp
⋮	⋮	⋮	⋮	⋮
1	R.H	Ahmed	Offered	1-6-2021 14:30
1	R.H	Ahmed	Allocated	1-6-2021 15:30
1	R.H	Ahmed	Started	1-6-2021 17:30
1	R.H	Ahmed	Suspended	1-6-2021 23:30
1	R.H	Ahmed	Started	1-6-2021 1:30
1	R.H	Nada	Offered	1-6-2021 2:30
1	R.H	Nada	Allocated	1-6-2021 03:30
1	R.H	Nada	Started	2-6-2021 18:30
1	R.H	Nada	Completed	2-6-2021 23:30
⋮	⋮	⋮	⋮	⋮

Sojourn Time

- Time between very first allocated/offered and last completed events.
- This measure applied only for activity dimension.



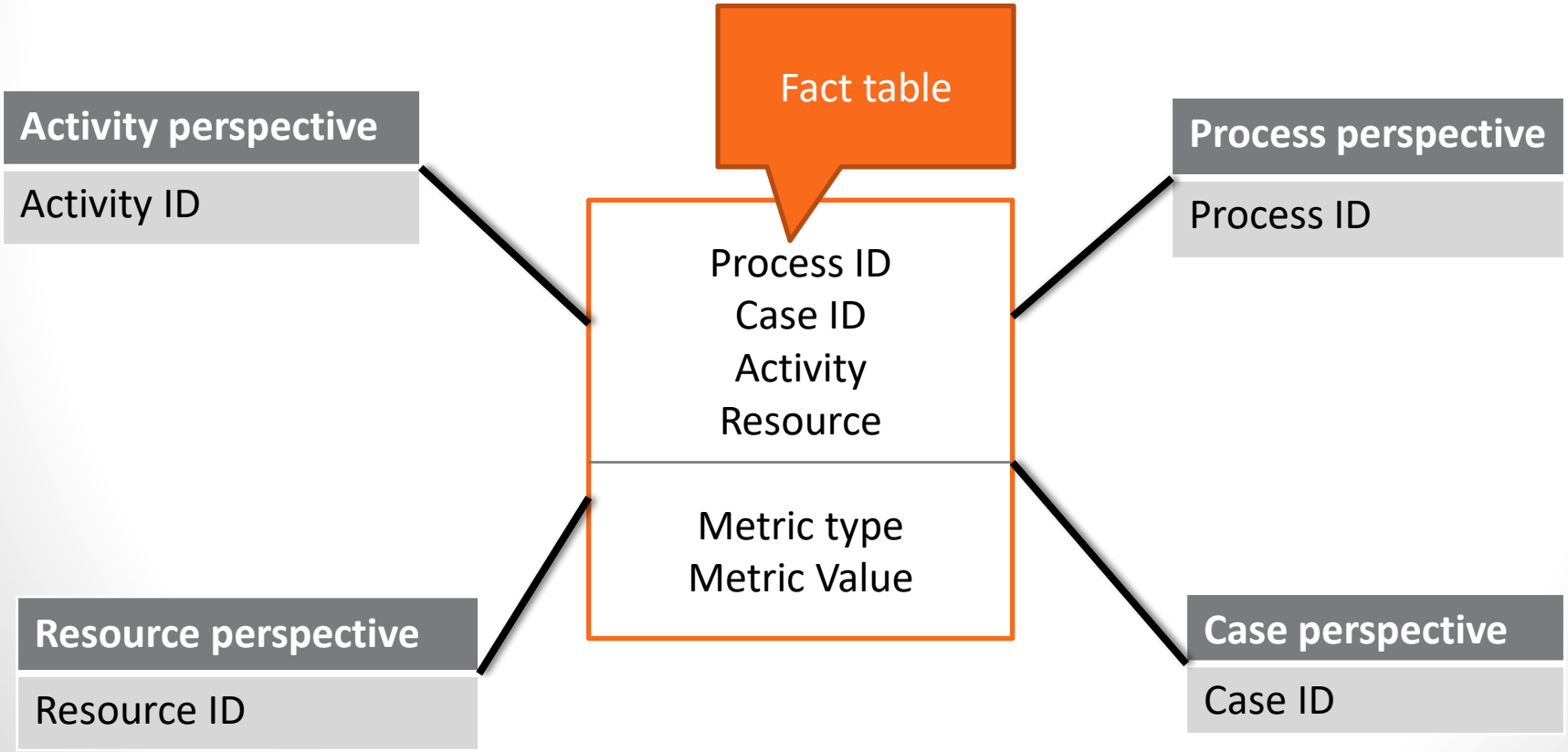
Sojourn Time

- **Example:**
  - Sojourn time for “Repair Hardware” in Case 1 is 33 hours

Case	Activity	Resource	Event type	Timestamp
⋮	⋮	⋮	⋮	⋮
1	R.H	Ahmed	Offered	1-6-2021 14:30
1	R.H	Ahmed	Allocated	1-6-2021 15:30
1	R.H	Ahmed	Started	1-6-2021 17:30
1	R.H	Ahmed	Suspended	1-6-2021 23:30
1	R.H	Ahmed	Started	1-6-2021 1:30
1	R.H	Nada	Offered	1-6-2021 2:30
1	R.H	Nada	Allocated	1-6-2021 03:30
1	R.H	Nada	Started	2-6-2021 18:30
1	R.H	Nada	Completed	2-6-2021 23:30
⋮	⋮	⋮	⋮	⋮

Raw Performance Measures

- is a tuple on the form
  - (Process id, Case id, Activity, Resource, Occurrence, Type, Start, End, Value)



Raw Performance Measures

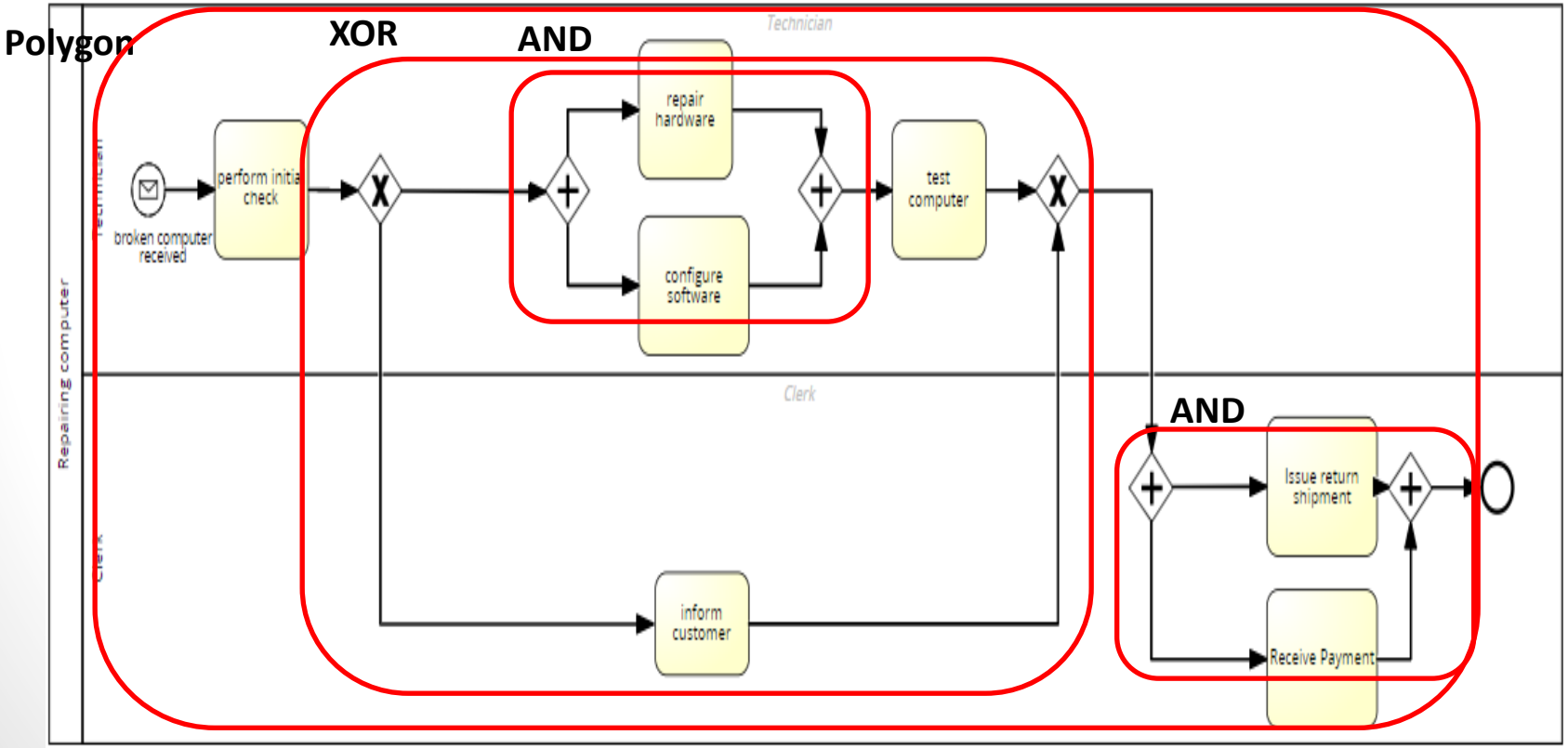
- Example: Obtain Effective time for Repair Hardware.

P.ID	CID	Activity	Resource	Occurrence	Event type	start		
	⋮	⋮	⋮		⋮	⋮		
1	1	R.H	Nada	1	Effective	1-6-2021 14:30	16-6-2021 17:30	7
1	1	R.H	Ahmed	1	Effective	1-6-2021 14:30	16-6-2021 17:30	9
1	1	R.H	Ahmed	1	Service	1-6-2021 14:30	16-6-2021 17:30	11
1	1	R.S	Mohamed	1	Waiting	1-6-2021 13:45	24-6-2021 14:30	5
1	1	R.S	Mona	1	Waiting	1-6-2021 13:45	24-6-2021 14:30	1
1	1	R.S	Mona	1	Service	20-6-2021 14:30	2-8-2021 14:30	22
1	1	R.S	Shady	2	Service	20-6-2021 14:30	2-8-2021 14:30	16
	⋮	⋮	⋮		⋮	⋮		

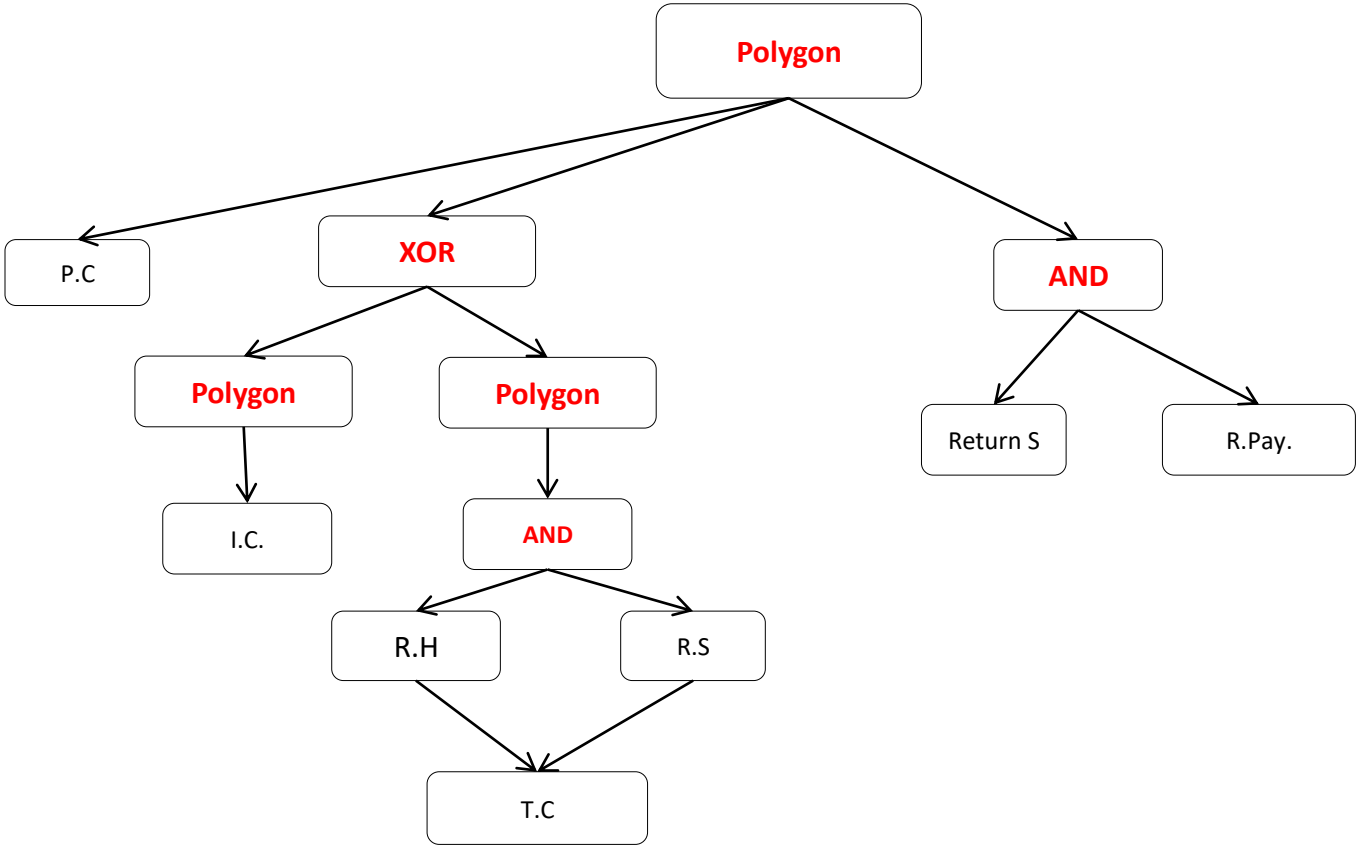
Effective time for R.H. is 16 hours

Refined Process Structure Tree

- RPST is based on the unique decomposition of a process model into fragments. Fragments, which are the decomposition result, are organized into a hierarchy according to the nesting relation.

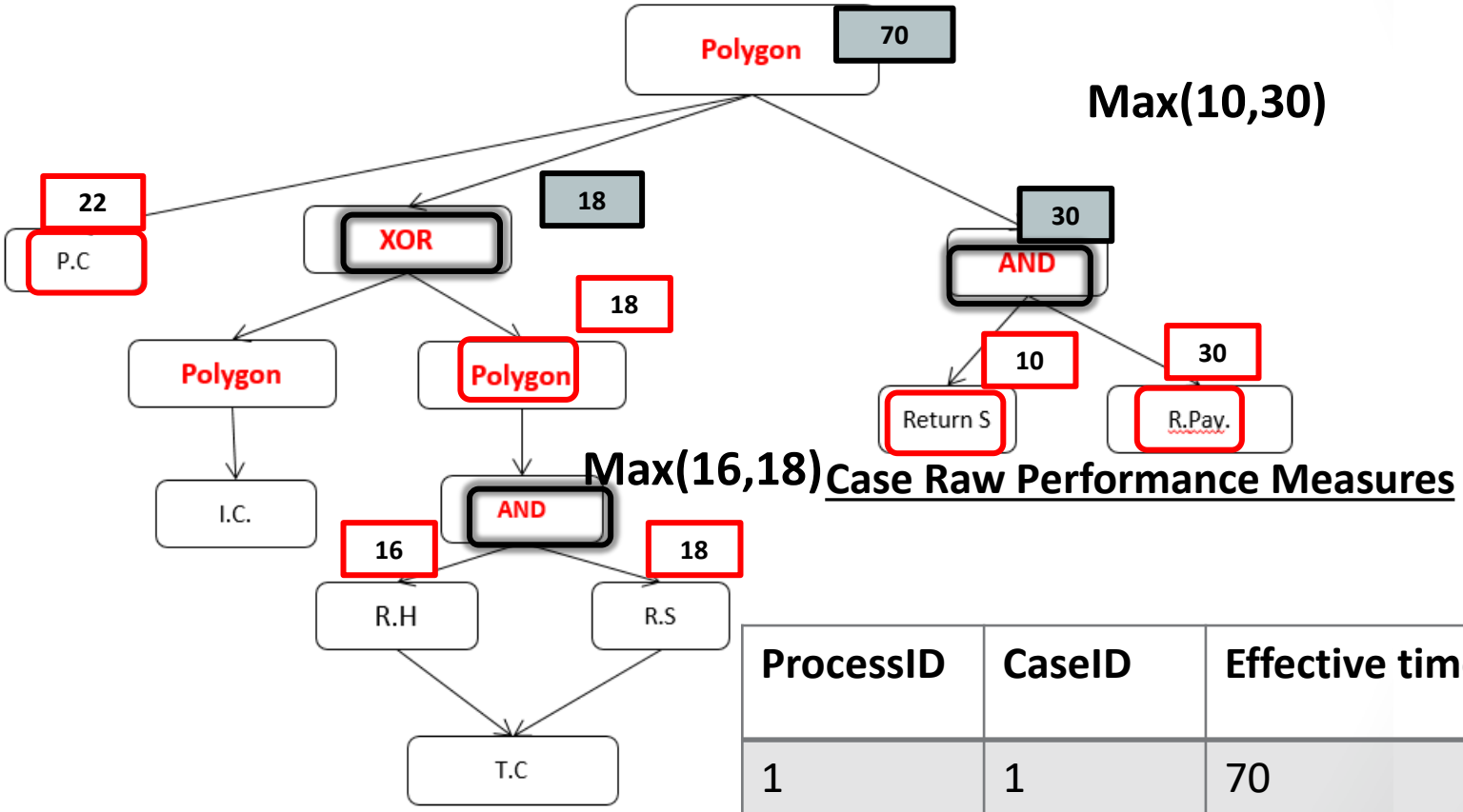


Refined Process Structure Tree



Example of Case Effective time

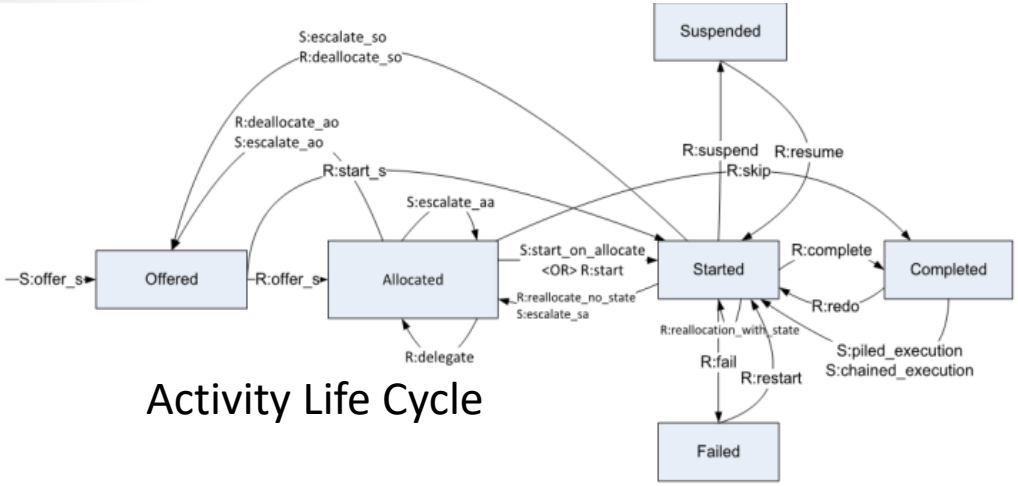
- Calculate Effective Time of Case 1 in process execution log:



Case Raw Performance Measures

ProcessID	CaseID	Effective time
1	1	70

ROWI Approach



Activity Life Cycle

Log contains overlapping activities

Event log

ROWI Approach

Log contains sequential activities

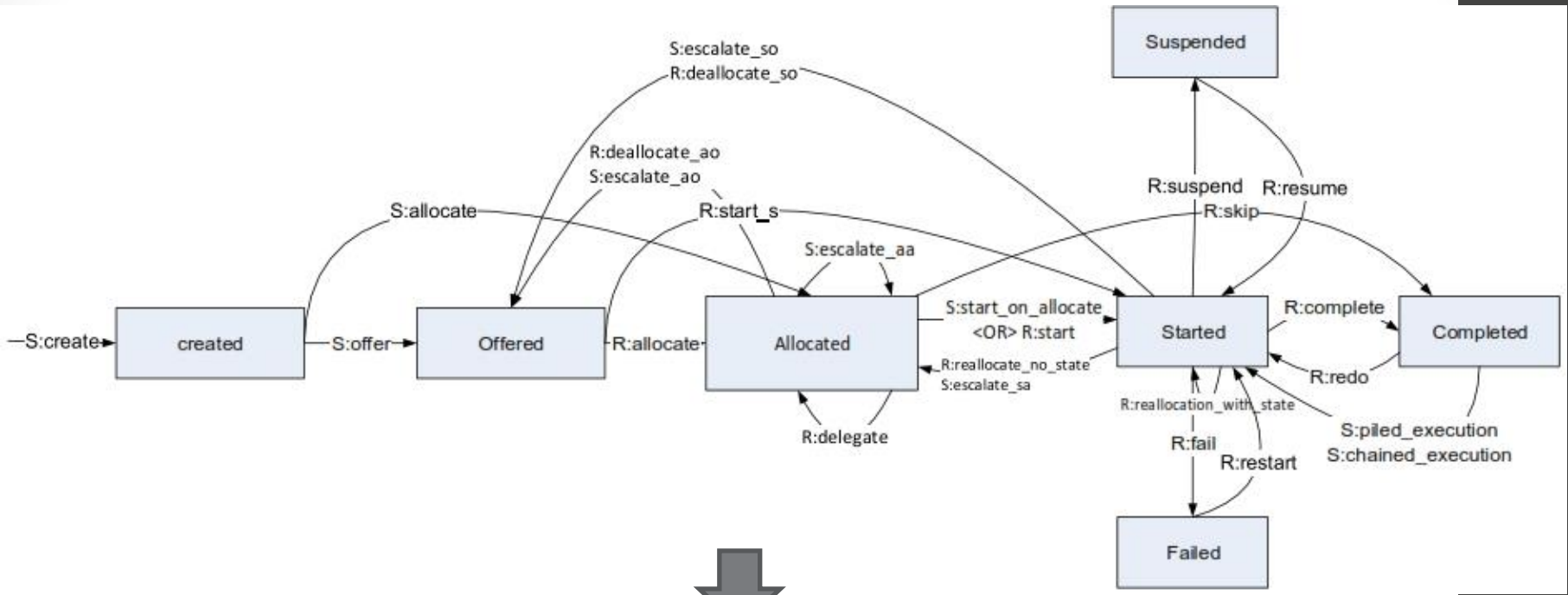
Enhanced log

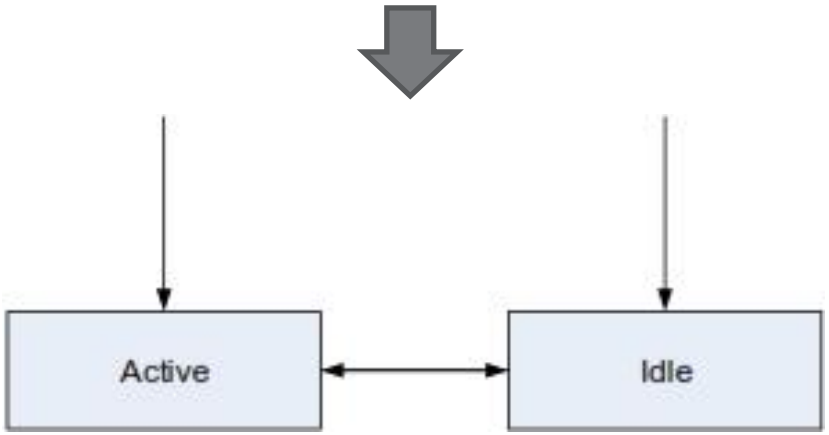
ROWI Approach

- In real logs, it often happens that some human resources appear to have more than one task active concurrently, thus resulting in human multitasking.

CID	Task	Resource	Event type	Timestamp
1	A	R1	Start	T1
2	B	R1	Start	T2
1	C	R1	Start	T3
1	A	R1	Complete	T4
2	B	R1	Complete	T5
1	C	R1	Complete	T6

Mapping event states into abstract states





Abstract Life Cycle

Compound Overlap

Active(A,t1) Active(B,t2) Idle (A,t3) Active(C,t4) Active(D,t5) Idle (B,t6) Idle (D,t7)

Relationship	Time
A o B	T2
B o C	T4
B o D	T5
C di D	T5

Queuing Pattern

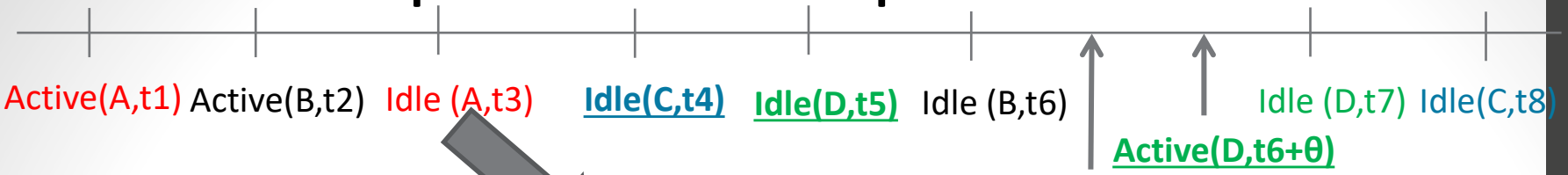
Active(A,t1) Active(B,t2) Idle (A,t3) Active(C,t4) **Idle(D,t5)** Idle (B,t6) Idle (D,t7) Idle(C,t8)

Active(D,t6+θ)

Relationship	Time
A o B	T2
B o C	T4
C di D	T5

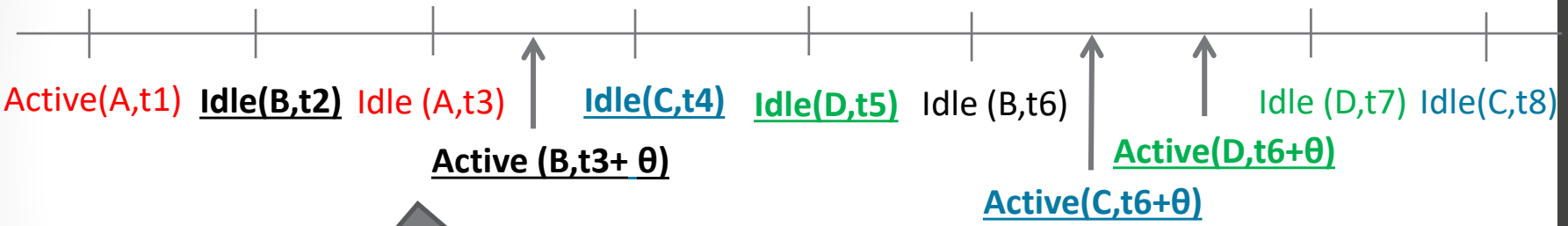
Queuing Pattern

Compound Overlap



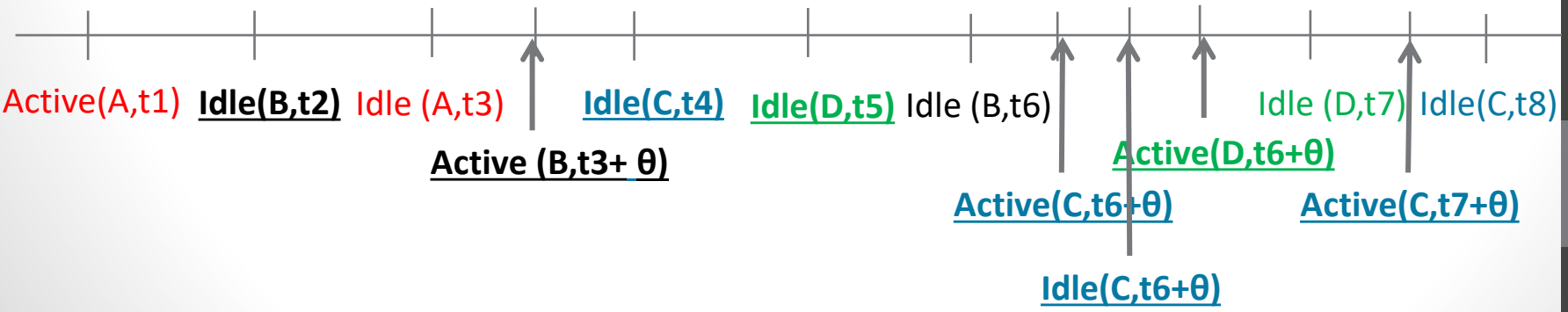
Relationship	Time
A o B	T2
C di D	T6

Queuing Pattern



Relationship	Time
C di D	T6

Stacking Pattern



Thank You